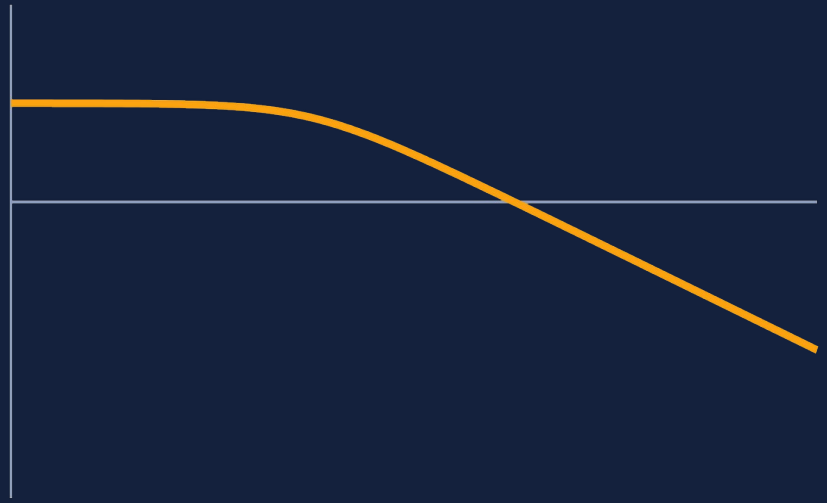
Lag, Lead and Lag-Lead Compensators



Control Systems (22EEG53)

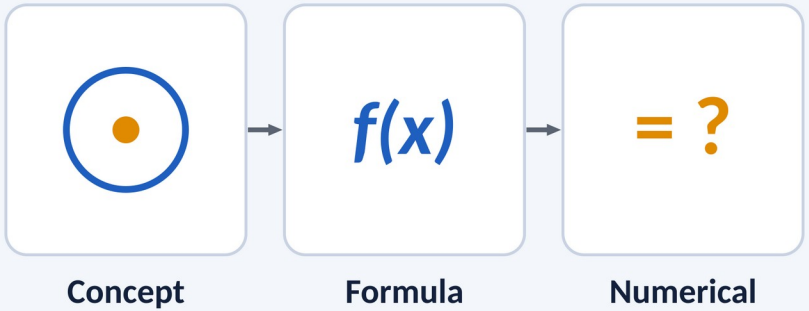
Unit 5: Compensators and Controllers

Topic: Introduction to compensators — lag, lead and lag-lead

Prof. Mitesh Astik

Learning Outcomes

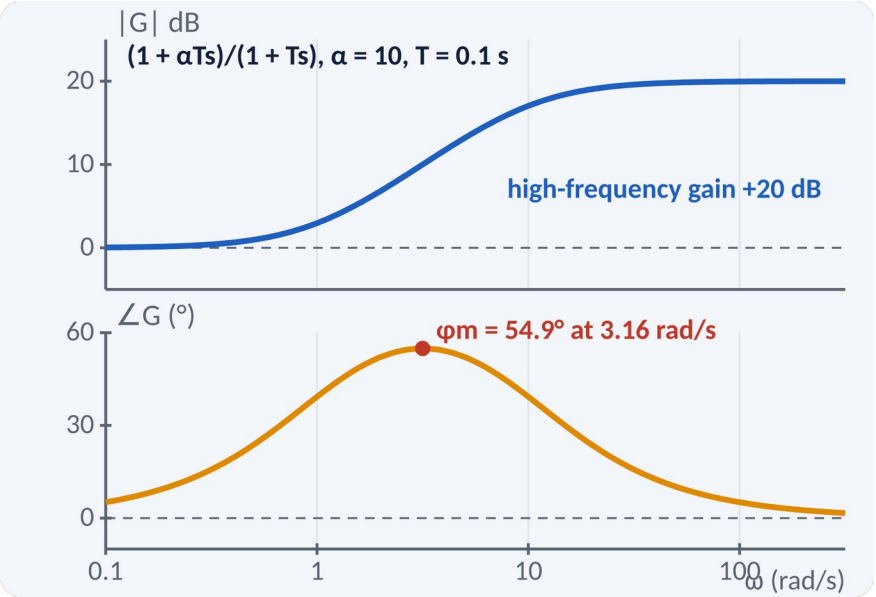
- Explain why compensation is needed.
- Write lag and lead transfer functions and their Bode features.
- Design a lead network for a phase-margin target.
- Select RC values for lag and lead networks.



Linked to CO5
Design the appropriate compensator and controller for given specifications.

Recap: What You Already Know

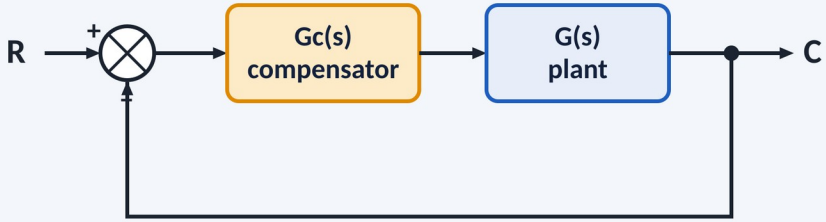
- Phase margin relates to damping (PM ≈ 100ζ).
- Gain K sets Kv and steady-state error.
- Bandwidth sets speed of response.
- Bode plot: pole −20 dB/dec, zero +20 dB/dec.



PM = 180° + ∠L(jωgc); Kv = lim s·L(s)
L(s) open-loop transfer function

Motivation: Gain Alone Is Not Enough

- Raise K: error falls but PM falls too.
- Specifications often conflict.
- Compensator adds phase or gain where needed.
- Simple, cheap, passive RC networks.



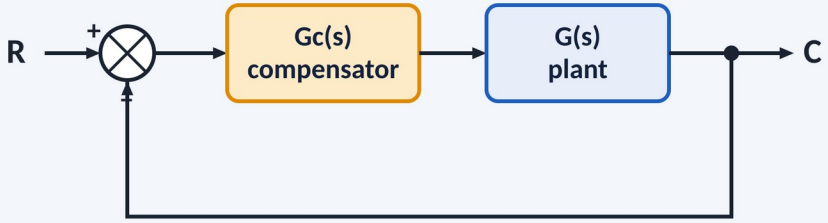
Series (cascade) compensation: loop gain L(s) = Gc(s)·G(s)

Goal: reshape L(s) to meet ess, ζ, ts, PM

Need: ess small and PM large and bandwidth high, at the same time

Series Compensation

- Compensator Gc(s) placed in the forward path.
- Open-loop becomes Gc(s)·G(s).
- Alternatives: feedback or feed-forward compensation.
- Lead, lag and lag-lead are the standard types.



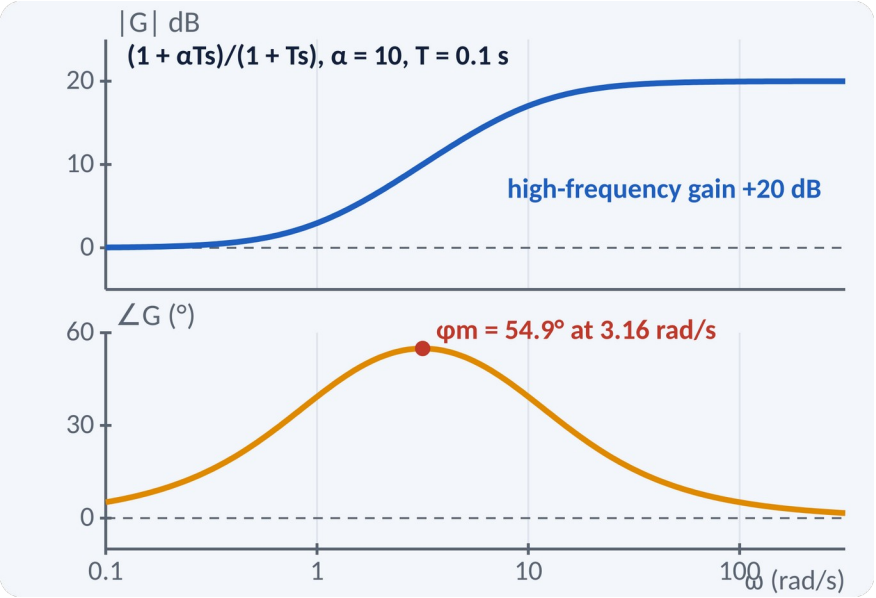
Series (cascade) compensation: loop gain L(s) = Gc(s)·G(s)

Goal: reshape L(s) to meet ess, ζ, ts, PM

L(s) = Gc(s)·G(s); Gc(s) = Kc·(s + z)/(s + p)

Lead Compensator

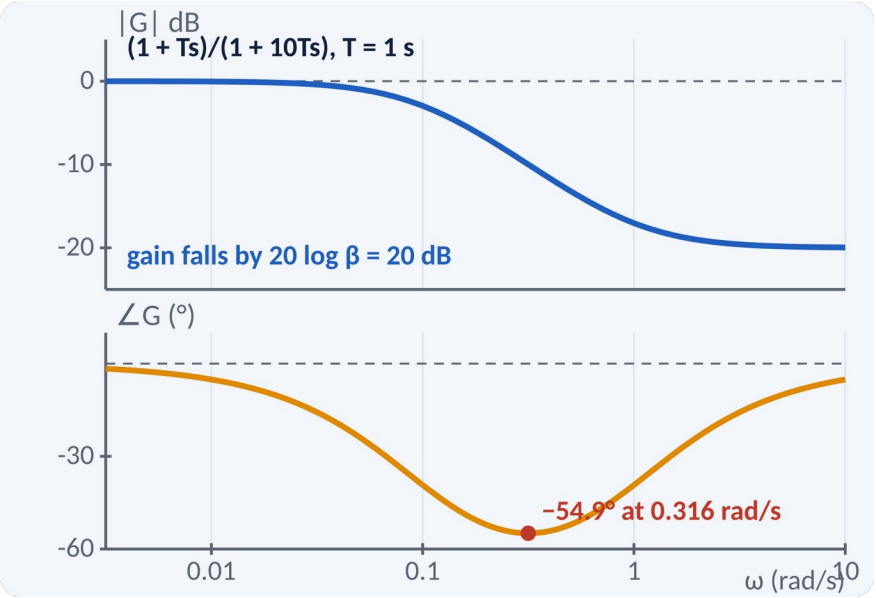
- Zero closer to the origin than the pole.
- Adds positive phase near ωm.
- Raises gain at high frequency.
- Widens bandwidth; faster, better damping.
- Noise at high frequency increases.



$G_c(s) = (1 + \alpha Ts)/(1 + Ts)$, $\alpha > 1$; $\varphi_m = \sin^{-1}[(\alpha - 1)/(\alpha + 1)]$ at $\omega_m = 1/(T\sqrt{\alpha})$
α lead ratio, T time constant (s)

Lag Compensator

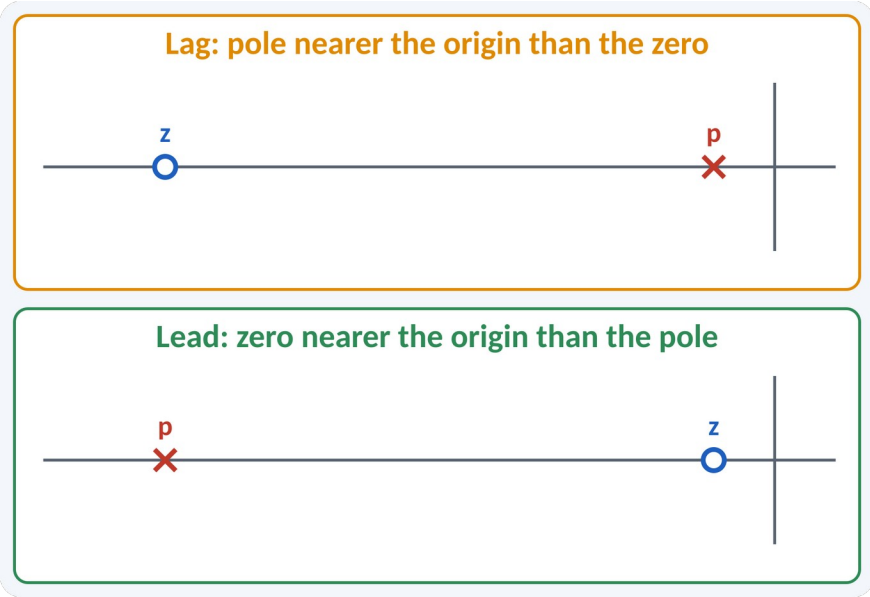
- Pole closer to the origin than the zero.
- Reduces gain at high frequency by 20 log β.
- Adds phase lag; place it well below ωgc.
- Allows a larger low-frequency gain.
- Reduces bandwidth; slower response.



$G_c(s) = (1 + Ts)/(1 + \beta Ts)$, $\beta > 1$; max lag $= \sin^{-1}[(\beta - 1)/(\beta + 1)]$ at $\omega = 1/(T\sqrt{\beta})$
β lag ratio, T (s)

Pole–Zero Pattern: Lag vs Lead

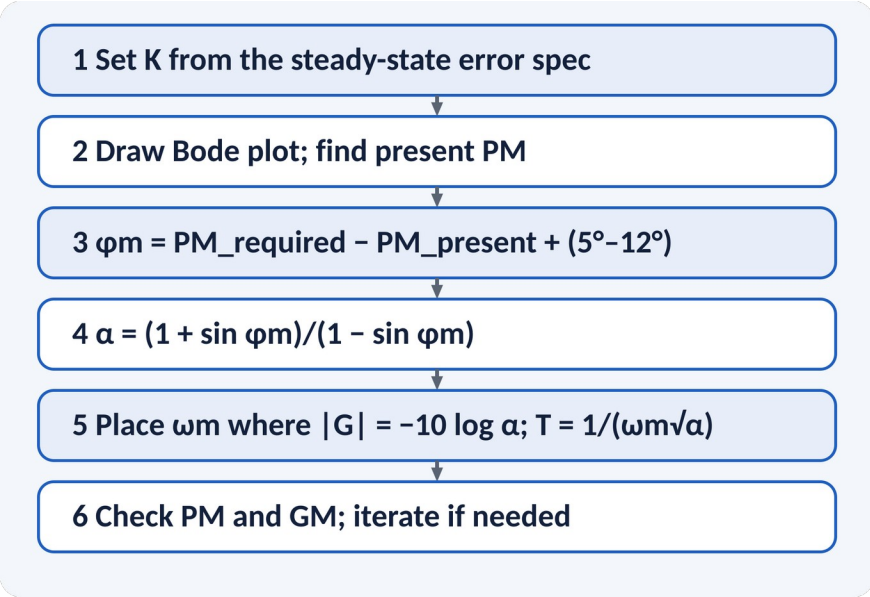
- Lead: zero at −1/(αT), pole at −1/T.
- Lag: zero at −1/T, pole at −1/(βT).
- Lag pole–zero pair sits close to the origin.
- Lead pair sits near the dominant poles.



Lead: |z| < |p|; Lag: |z| > |p|

Lead Design Procedure (Bode)

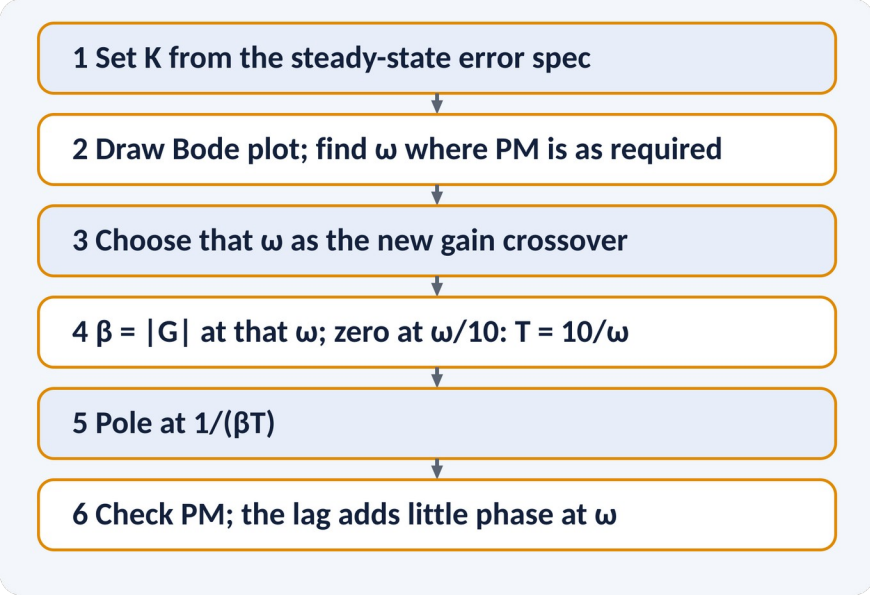
- Fix K for the steady-state specification.
- Compute the phase lead needed.
- Add 5°–12° to cover the gain shift.
- Compute α and place ωm.
- Verify margins and iterate.



α = (1 + sin φm)/(1 − sin φm); T = 1/(ωm√α); gain at ωm = +10 log α dB

Lag Design Procedure (Bode)

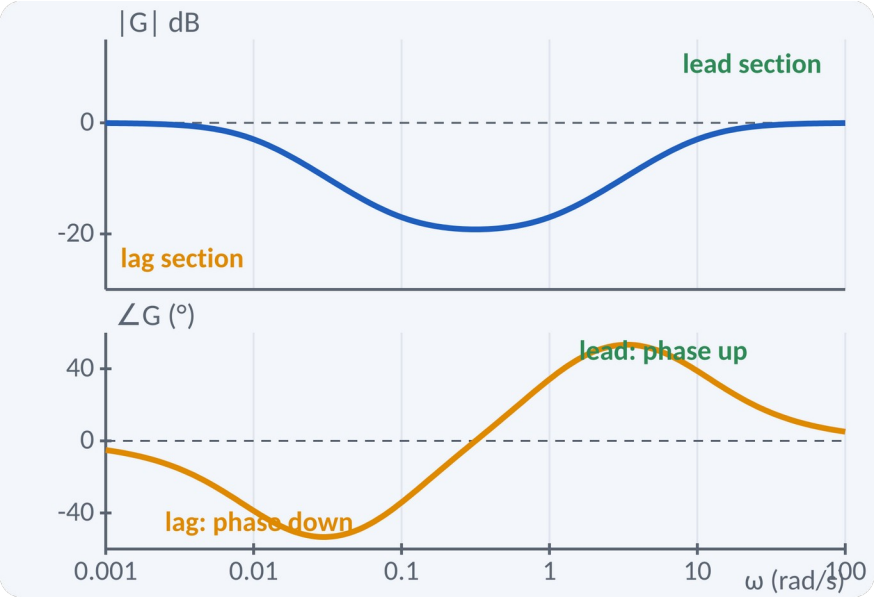
- Fix K for the steady-state specification.
- Find where the plant phase gives the required PM.
- Make that frequency the new crossover.
- β from the plant gain there; zero a decade below.
- Check the final margins.



$\beta = |G(j\omega'_{gc})|$; $1/T = \omega'_{gc}/10$; pole at $1/(\beta T)$

Lag–Lead Compensator

- Combines a lag section and a lead section.
- Lag improves steady-state accuracy.
- Lead improves speed and damping.
- Used when both are needed.
- Lag corners lie below, lead corners above.



$$G_c(s) = [(1 + T_1 s)/(1 + \beta T_1 s)] \cdot [(1 + T_2 s)/(1 + T_2 s/\beta)]$$

RC Realisations and Comparison

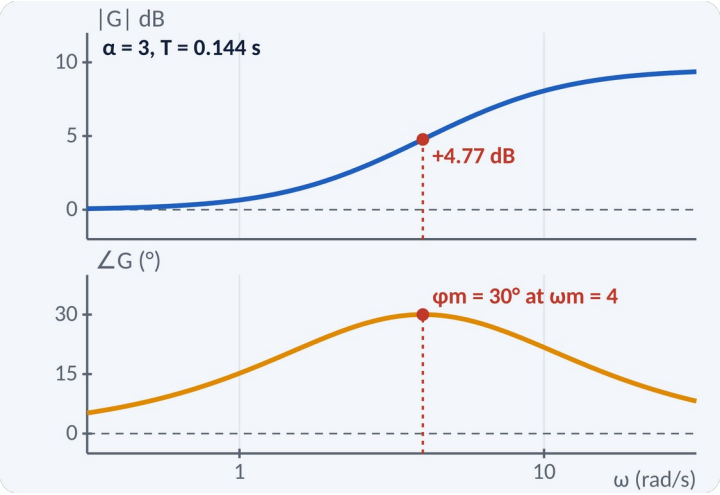
- Lag network: R1 in series, R2–C to ground.
- Lead network: R1‖C in series, R2 to ground.
- Lead network loses gain 1/α; use an amplifier.
- Lab: build and test both networks.

	Lead	Lag
Form	(1+αTs)/(1+Ts)	(1+Ts)/(1+βTs)
Phase	adds lead	adds lag
Bandwidth	increases	decreases
Speed	faster	slower
Steady-state	little change	improves
Noise	more	less

Lag: β = (R1 + R2)/R2, T = R2C; Lead: α = (R1 + R2)/R2, αT = R1C

Solved Example 1: Lead Network

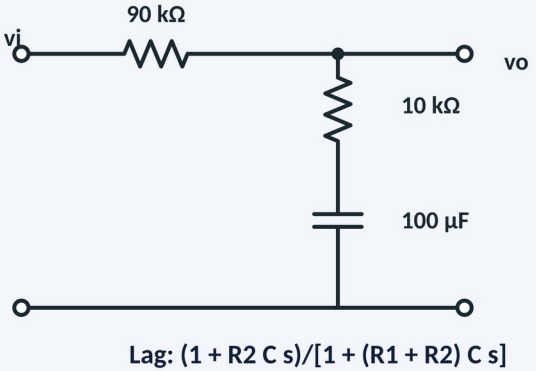
- A loop has PM = 20°. Required PM = 45°; use 5° margin. Find α and T for ωm = 4 rad/s.
- 1. φm = 45° − 20° + 5° = 30°.
- 2. α = (1 + sin 30°)/(1 − sin 30°) = 1.5/0.5 = 3.
- 3. T = 1/(ωm√α) = 1/(4 × 1.732) = 0.144 s.
- 4. Zero at 1/(αT) = 2.31, pole at 1/T = 6.93 rad/s.
- 5. Gain at ωm = 10 log 3 = 4.77 dB.



Answer: Gc(s) = (1 + 0.433s)/(1 + 0.144s); zero 2.31, pole 6.93 rad/s

Solved Example 2: RC Lag Network

- Design an RC lag with β = 10 and T = 1 s. Find R1, R2, C and the corner frequencies.
- 1. T = R2C = 1 s: choose R2 = 10 kΩ, C = 100 µF.
- 2. β = (R1 + R2)/R2 = 10 → R1 = 9R2 = 90 kΩ.
- 3. Zero at 1/T = 1 rad/s; pole at 1/(βT) = 0.1 rad/s.
- 4. High-frequency attenuation = 20 log 10 = 20 dB.
- 5. Maximum lag: $\sin^{-1}(9/11) = 54.9°$ at 1/(T√β) = 0.316 rad/s.



Lag: (1 + R2 C s)/[1 + (R1 + R2) C s]

Answer: R1 = 90 kΩ, R2 = 10 kΩ, C = 100 µF; Gc(s) = (1 + s)/(1 + 10s)

Common Mistakes and Exam Tips

Common mistakes

- Using lag when more phase lead is needed.
- Placing the lag corner close to ωgc.
- Forgetting the 1/α loss in a passive lead network.
- No safety margin in φm.
- Not rechecking margins after the shift in crossover.

Exam tips

- State the problem first: speed or accuracy?
- Lead for speed/damping, lag for accuracy.
- Add 5°–12° to φm for the lead.
- Verify with the Bode plot or MATLAB.

Summary

- Compensator = pole–zero pair in series with the plant.
- Lead: adds phase, raises bandwidth, speeds up.
- Lag: cuts high-frequency gain, improves accuracy.
- Lag–lead gives both effects.
- $\alpha = (1 + \sin \varphi m)/(1 - \sin \varphi m)$; $\omega m = 1/(T\sqrt{\alpha})$.

	Lead	Lag
Form	$(1+\alpha Ts)/(1+Ts)$	$(1+Ts)/(1+\beta Ts)$
Phase	adds lead	adds lag
Bandwidth	increases	decreases
Speed	faster	slower
Steady-state	little change	improves
Noise	more	less

Lead: zero before pole; Lag: pole before zero

Quick Quiz

Q1. A lead compensator mainly improves:
(a) steady-state error (b) phase margin and speed (c) noise (d) gain margin only

Ans: (b)

Q2. Maximum lead for α = 3 is:
(a) 19.5° (b) 30° (c) 45° (d) 60°

Ans: (b)

Q3. A lag network reduces high-frequency gain by:
(a) 20 log α (b) 20 log β (c) 10 log β (d) zero

Ans: (b)

Q4. In a lag network the pole is:
(a) nearer to origin than zero (b) farther (c) at origin (d) at infinity

Ans: (a)

Q5. For lead with α = 10, ωm occurs at:
(a) 1/T (b) 1/(T√α) (c) α/T (d) 1/(αT)

Ans: (b)

References

1. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3, 4, 5 textbook)

2. **J. J. D'Azzo and C. H. Houpis, Feedback Control System Analysis and Synthesis**
   4th Ed., McGraw Hill International, 1999 (reference, Unit 5)

3. **M. Gopal, Control Systems Principles and Design**
   4th Ed., TMH, 1999 (reference, Units 4, 5)

Next lecture: P, PI and PID controllers